

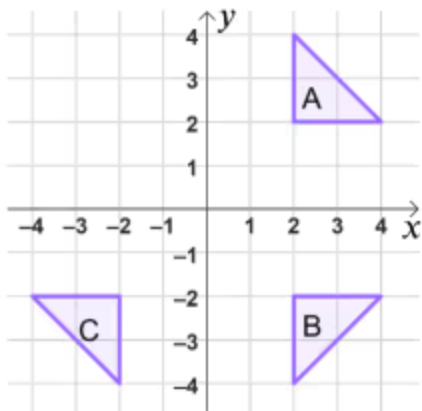


Problem solving with transformations

- 1 The perimeter and area are the same after a translation, rotation or reflection because the object and image are congruent. Fill in the blank
- 2 A shape is rotated 90° anti-clockwise about $(2, 7)$. Which of the following rotations would result in the image being moved to the same position as the original object? Tick **2** correct answers
- ☒ Rotation 90° clockwise about $(2, 7)$
 - ☐ Rotation 90° clockwise about $(7, 2)$
 - ☒ Rotation 270° anti-clockwise about $(2, 7)$
- 3 The coordinate (a, b) has been translated by $\begin{pmatrix} 4 \\ 3 \end{pmatrix}$. The image of this point will have the coordinate _____. Tick **1** correct answer
- ☐ $(4a, 3b)$
 - ☐ $(a4, b3)$
 - ☒ $(a + 4, b + 3)$
 - ☐ $(a - 4, b - 3)$



- 4 Shape A is reflected in the x -axis to create shape B. Shape B is then reflected in the y -axis to create shape C. Which transformations would map shape C onto shape A? Tick **2** correct answers



- ☒ A rotation 180° about the origin.
- ☒ A reflection in the line that passes through $(-4, 4)$ and $(3, -3)$
- ☐ A translation by $\begin{pmatrix} 4 \\ 4 \end{pmatrix}$
- ☐ A reflection in the line that passes through $(4, 4)$ and $(-2, -2)$

- 5 A shape with perimeter 16 cm is enlarged by scale factor 3. The perimeter of the image is 48 cm cm. Fill in the blank

- 6 The point $A(p + 3, q + 5)$ has the image $A'(p + 1, q + 8)$. Which vector has it been translated by? Tick **1** correct answer

☐ $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$

☒ $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$

☐ $\begin{pmatrix} 2 \\ -3 \end{pmatrix}$

☐ $\begin{pmatrix} -2p \\ 3q \end{pmatrix}$

